

SinusCare

An Uncertainty-Aware Adaptive Machine Learning Framework for Personalized Sinusitis Risk and Recurrence Assessment

Purpose: IEEE conference research-project blueprint

Recommended direction: Build an adaptive, explainable and longitudinal screening framework rather than a basic sinusitis classifier.

1. Executive Recommendation

I recommend going with **SinusCare: An Uncertainty-Aware Adaptive Machine Learning Framework for Personalized Sinusitis Risk Assessment**. The core contribution should be an interactive system that starts with a small set of patient inputs, estimates risk and uncertainty, selects the most informative next information when uncertainty is high, and then combines the final prediction with longitudinal history, environmental context and explainability.

Do **not** position the paper as the first AI system for sinusitis, the first recurrence model, or the first model using air pollution. Existing research already covers AI diagnosis/prognosis in rhinology and has specifically used particulate-matter exposure and machine learning for chronic rhinosinusitis with nasal polyps recurrence prediction. The novelty should instead be the proposed **adaptive, uncertainty-aware, personalized framework** and its experimental evaluation.

2. Recommended Research Title

Primary title: SinusCare: An Uncertainty-Aware Adaptive Machine Learning Framework for Personalized Sinusitis Risk Assessment

Alternative: SinusCare: An Adaptive and Explainable Framework for Personalized Sinusitis Risk and Recurrence Assessment

Why this title: It makes the technical contribution visible—adaptive information acquisition and uncertainty—not merely the healthcare application.

3. Research Problem

Sinusitis and chronic rhinosinusitis can present with variable symptoms and may recur. Existing AI work in rhinology has heavily explored imaging, endoscopy, diagnosis, phenotyping and prognosis. However, clinical/patient-reported predictive modeling remains exploratory, and recent reviews identify challenges including external validation, workflow integration, interpretability, interoperability, privacy and equitable performance.

The proposed study asks: **Can an adaptive screening framework use only the most informative available patient information to provide useful sinusitis risk assessment, while incorporating longitudinal history and environmental context and communicating prediction uncertainty and explanations?**

4. Core Novelty

Contribution	What is new in your project	How to validate it
Adaptive feature acquisition	The model chooses additional questions/features when the current prediction is uncertain.	Compare adaptive vs fixed full-question screening on performance and number of inputs.

Contribution	What is new in your project	How to validate it
Uncertainty-aware prediction	Model disagreement/calibration influences whether more information is requested.	Measure calibration, uncertainty and error rates; evaluate whether uncertain cases benefit from more inputs.
Longitudinal personalization	Use previous episodes and symptom trends rather than treating each assessment as independent.	Compare current-only vs history-aware models.
Environmental context	Align patient records with environmental variables such as PM2.5, PM10, AQI, temperature and humidity where data permits.	Ablation study: clinical/history model vs clinical/history/environment model.
Explainability	Provide patient-specific feature contributions instead of only a risk label.	Use SHAP/permutation importance and assess consistency/interpretability.

5. System Overview

Pipeline: Initial patient data → baseline prediction → uncertainty estimation → adaptive selection of the next informative feature → updated prediction → final risk → recurrence/trend analysis → explainable AI → dashboard/report.

Important: The system is a screening/decision-support prototype, not a diagnostic device. It must not claim to diagnose sinusitis or prescribe treatment.

Conceptual flow:

Patient symptoms/history → preprocessing → baseline model → risk + uncertainty → adaptive question/feature selection → updated model → personalized risk → recurrence/trend → SHAP explanation → longitudinal dashboard.

6. Main Functional Modules

Patient Profile: Age, sex and relevant non-sensitive profile variables.

Symptom Assessment: Nasal congestion, discharge, facial pressure/pain, headache, reduced smell, cough, postnasal drip, fever, fatigue, duration and severity.

Medical/History Module: Previous episodes, recurrence count, allergy history, asthma, nasal polyps and relevant prior history where available.

Environmental Module: AQI, PM2.5, PM10, temperature, humidity and other available contextual variables.

Adaptive Screening Engine: Selects additional information when current prediction uncertainty is high.

Current Risk Model: Estimates screening risk from available information.

Recurrence/Trend Model: Uses longitudinal records to estimate recurrence tendency or risk trend when sufficient data exist.

Explainable AI: Displays patient-specific factors influencing the model output.

Longitudinal Dashboard: Shows symptom/risk history and environmental context over time.

Report Module: Produces a concise screening report with limitations and clinician-review disclaimer.

7. Adaptive Screening Method

The most research-oriented component is adaptive feature acquisition. The system does not require every feature at the beginning.

Step 1: Collect a small initial feature set (for example, key symptoms, duration and prior episode information).

Step 2: Produce a preliminary prediction and uncertainty estimate.

Step 3: If confidence is insufficient, estimate which missing feature would provide the greatest expected information gain or performance improvement.

Step 4: Request/obtain that feature and update the prediction.

Step 5: Stop when a predefined confidence/utility criterion is reached or a maximum number of questions is exhausted.

Research metric: Compare predictive performance against the number of features/questions required.

8. Uncertainty-Aware Design

A healthcare screening system should not behave as if every prediction is equally reliable. Possible approaches include probability calibration, ensemble disagreement, bootstrap uncertainty, or conformal prediction depending on the available expertise and dataset size.

Example logic: if several models agree and calibration is good, return the screening result. If models disagree or uncertainty is high, request another informative feature rather than presenting an overconfident result.

Key research question: Does uncertainty-guided information acquisition improve the trade-off between prediction quality and data-collection burden?

9. Longitudinal Personalization

Each assessment should be stored with a timestamp. Over multiple assessments, the framework can analyze symptom severity, episode frequency, time between episodes and environmental context.

A simple first version can use engineered temporal features such as number of episodes in the previous N days, days since previous episode, recent symptom severity trend and rolling environmental exposure. A more advanced version can investigate temporal models only if enough longitudinal data are available.

Do not claim that a future recurrence probability is clinically valid unless you have an appropriate time-to-event/longitudinal dataset and proper validation.

10. Environmental Integration

Potential environmental features include AQI, PM2.5, PM10, temperature, humidity and NO2, with pollen added only if reliable data are available. Historical environmental data can be aligned by date and geographic region when the underlying patient dataset supports this.

The model should describe these variables as **associated factors** unless causal inference is specifically designed and justified. Avoid statements such as 'pollution caused your sinusitis.'

Experimental comparison: clinical-only model vs clinical + history vs clinical + history + environment.

11. Explainable AI

Use SHAP as the primary explanation method, with permutation importance as a secondary global check. For each prediction, show the most influential factors and their direction/relative contribution.

Example: 'The model's current estimate was influenced most by symptom duration, previous episodes, nasal congestion and recent environmental exposure.' This is an explanation of the model—not a medical conclusion.

12. Dataset Strategy

Challenge: A single public dataset containing symptoms, detailed history, longitudinal recurrence, geographic information and matched daily environmental exposure may be difficult to obtain.

Recommended approach: First locate a public clinical/sinusitis dataset and build a baseline. Then identify a compatible environmental dataset. If you create a prototype longitudinal dataset, clearly label it as synthetic, simulated, or prospectively collected with appropriate consent/ethics procedures.

Do not: fabricate patient records, merge incompatible datasets without justification, or report synthetic data as real clinical evidence.

13. Feature Groups

Group	Example features
Symptoms	Congestion, discharge, facial pressure/pain, headache, smell reduction, cough, postnasal drip, fever, fatigue, duration, severity.
History	Previous episodes, episode count, time since last episode, allergy history, asthma, nasal polyps, prior treatment/surgery if available.
Exposure	Smoking exposure, dust, occupational exposure, indoor ventilation where reliably measured.
Environment	AQI, PM2.5, PM10, temperature, humidity, NO2, pollen if available.
Temporal	Recent trend, rolling averages, episode frequency, time between episodes, recent exposure windows.

14. Machine-Learning Methodology

Baseline models: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, XGBoost/LightGBM where appropriate.

Preprocessing: missing-value handling, categorical encoding, scaling where required, outlier policy, class-imbalance handling, and leakage checks.

Feature selection: use a training-fold-only procedure. Do not select features using the full dataset before splitting.

Validation: stratified cross-validation for independent samples; temporal or grouped validation for longitudinal/repeated-patient data where appropriate.

Calibration: assess whether predicted probabilities correspond to observed frequencies; do not report raw model probabilities as clinically meaningful risk without calibration.

15. Experiments for the IEEE Paper

ID	Experiment	Purpose
E1	Symptoms-only baseline	Establish a transparent baseline.
E2	Symptoms + history	Test value of longitudinal/personal history.
E3	Symptoms + history + environment	Test incremental environmental value.
E4	Fixed full-feature model vs adaptive model	Test whether similar performance can be achieved with fewer inputs.
E5	Uncertainty-aware vs non-uncertainty adaptive selection	Test whether uncertainty improves feature selection.
E6	Explainability analysis	Check whether dominant model factors are stable and interpretable.

16. Evaluation Metrics

- Sensitivity/Recall
- Specificity
- Precision
- F1-score
- ROC-AUC
- PR-AUC for class imbalance
- Confusion matrix
- Calibration/Brier score
- Number of features/questions required
- Performance vs data-collection burden

For screening, do not rely on accuracy alone. Sensitivity, specificity, calibration and error analysis are particularly important.

17. Research Hypotheses

1. Adding patient history improves risk prediction over symptom-only models.
2. Adding environmental context provides measurable incremental predictive value over clinical/history features.
3. Uncertainty-guided adaptive feature acquisition can reduce the number of required inputs while maintaining acceptable predictive performance.
4. Longitudinal information improves personalized risk/trend estimation compared with single-visit assessment.
5. Explainable AI can identify consistent, patient-specific factors contributing to model predictions.

18. Proposed Research Gap

Recent literature shows substantial AI activity in rhinology, particularly automated CT analysis, endoscopy, phenotyping and prognosis. A 2026 state-of-the-art review describes integrated clinical/patient-reported predictive modeling as still exploratory and highlights the need for prospective validation, workflow integration, transparency and equitable performance. This supports investigating an interactive, patient-centered framework rather than another static image classifier.

Proposed gap statement: Existing work does not fully address the combination of uncertainty-aware adaptive information acquisition, longitudinal personalization, environmental context and patient-specific explanation within a single lightweight screening workflow. The proposed study therefore evaluates whether this integrated framework can improve the efficiency and transparency of sinusitis risk assessment.

19. What Counts as Your Novel Contribution

Strong claim: 'We propose and experimentally evaluate an uncertainty-aware adaptive framework for personalized sinusitis risk assessment.'

Stronger if supported: 'The framework reduces the number of required inputs while maintaining comparable predictive performance to a fixed full-feature baseline.'

Strongest if supported by data: 'We demonstrate that uncertainty-guided feature acquisition provides a better performance–data burden trade-off than fixed or random feature acquisition.'

Do not claim novelty merely because you used Random Forest, XGBoost, SHAP, AQI, or a web dashboard. Those are components; the research contribution is the validated method and evidence.

20. System Architecture / Technology Stack

Frontend: React.js or Next.js + Bootstrap/Tailwind + Chart.js/Recharts.

Backend: FastAPI/Flask for ML service, or Node.js/Express plus a Python ML service.

Database: PostgreSQL or MongoDB. PostgreSQL is preferable if you need structured longitudinal records and analytical queries.

ML: Python, Pandas, NumPy, Scikit-learn, XGBoost/LightGBM, SHAP.

Deployment: Containerized API + database + frontend; keep identifiable health data out of public demo deployments.

21. User Workflow

1. User creates a profile.
2. Completes a short initial symptom assessment.
3. System produces preliminary risk and uncertainty.
4. If uncertainty is high, the adaptive engine selects additional questions/features.
5. Final screening estimate is generated.
6. SHAP-based explanation is shown.
7. User can record future episodes and symptoms.
8. Longitudinal dashboard tracks trends and environmental context.
9. Report explicitly states that the result is not a diagnosis.

22. Safety, Ethics and Privacy

- Use the system for screening/research support, not diagnosis or treatment.
- Do not provide medication or treatment recommendations from the ML model.
- Use appropriate consent/ethics approval if collecting real patient data.
- Minimize personally identifiable information.
- Protect location and health data; avoid storing precise location unless necessary.
- Report demographic/subgroup performance where sample size permits.
- Clearly separate real, public, synthetic and simulated data.
- Use a clinician-reviewed interpretation if the project is ever evaluated in a clinical setting.

23. Limitations to State in the Paper

- Longitudinal sinusitis datasets with matched environmental exposure may be difficult to obtain.
- Self-reported symptoms can introduce reporting bias.
- Environmental associations do not prove individual-level causation.
- A model trained on one population may not generalize to another.
- Recurrence prediction requires appropriate longitudinal outcomes and validation.
- A student prototype cannot establish clinical safety or clinical effectiveness.
- External and prospective validation are required before real-world clinical use.

24. Future Work

Potential extensions include pollen exposure, indoor air-quality sensors, wearable-derived sleep/activity data, clinician feedback, endoscopy/image analysis, multimodal learning, federated learning and multicenter prospective validation. These should remain future work unless the required datasets and approvals are available.

25. IEEE Paper Structure

1. Abstract
2. Index Terms
3. I. Introduction
4. II. Related Work / Literature Review
5. III. Research Gap and Contributions
6. IV. Proposed SinusCare Framework
7. V. Dataset and Preprocessing
8. VI. Adaptive and Uncertainty-Aware Methodology
9. VII. Experimental Setup
10. VIII. Results and Comparative Analysis
11. IX. Explainability and Error Analysis
12. X. Discussion
13. XI. Limitations, Ethics and Privacy
14. XII. Conclusion and Future Work
15. References

26. Short Abstract for Proposal Form

SinusCare is an explainable AI-based framework for personalized sinusitis risk and recurrence assessment using patient symptoms, medical history and environmental conditions. The proposed system uses uncertainty-aware adaptive screening to select informative inputs, combines longitudinal patterns with environmental context, and provides interpretable risk estimates for early screening and monitoring. The framework is intended as a research/decision-support prototype and not as a replacement for clinical diagnosis.

27. Final Recommendation

Go with SinusCare, but make the research contribution adaptive and uncertainty-aware. Do not build the project around a generic sinusitis classifier. Your minimum publishable research core should be: (1) baseline symptom model, (2) history-aware model, (3) environment-aware model, (4) adaptive feature acquisition, (5) uncertainty/calibration analysis, and (6) explainable predictions.

The paper becomes strong if your experiments show a measurable benefit—especially a better **performance versus number-of-inputs** trade-off. If the adaptive method does not improve anything, that is also a legitimate research result, provided the experiment is rigorous and honestly reported.

28. Literature Evidence Used for Topic Selection

Wang et al., “Individual exposure of ambient particulate matters and eosinophilic chronic rhinosinusitis with nasal polyps: Dose-Response, mediation effects and recurrence prediction,” *Environment International*, 2023, DOI: 10.1016/j.envint.2023.108031.

The study used 1,086 CRSwNP patients from nine hospitals and machine learning for recurrence prediction using PM exposure, inflammatory and demographic variables.

Hack and Takashima, "Artificial Intelligence in Rhinology: A State-of-the-Art Review of Clinical Readiness and Implementation Pathways," *Otolaryngology–Head and Neck Surgery*, 2026. The review describes current AI domains in rhinology and identifies gaps including prospective validation, workflow integration, transparency, equity and the exploratory state of integrated clinical/patient-reported predictive modeling.

Liu et al., "Artificial intelligence optimizes the standardized diagnosis and treatment of chronic sinusitis," *Frontiers in Physiology*, 2025, DOI: 10.3389/fphys.2025.1522090. The review summarizes AI applications in CRS diagnosis, prognosis and treatment and discusses standardization, interpretability, collaboration and privacy challenges.

Alanazi et al., "Diagnostic Accuracy of Artificial Intelligence Models in Computed Tomography Interpretation of Chronic Rhinosinusitis: A Systematic Review," *Ear, Nose & Throat Journal*, 2026, DOI: 10.1177/01455613261423741. The review reports substantial existing work on AI-based CT interpretation and highlights methodological limitations and limited external validation.

Chaudhary and Dahan, "Artificial Intelligence-Enhanced Cone-Beam Computed Tomography for the Diagnostic Evaluation of Chronic Rhinosinusitis: A Systematic Review," *Cureus*, 2025, DOI: 10.7759/cureus.94552. The review identifies numerous imaging-AI studies and emphasizes the need for standardized protocols, patient-centered outcomes and multicenter prospective validation.

Important: This blueprint is a research-planning document. Literature claims should be rechecked and expanded during the formal systematic literature review before IEEE submission. The project should not be presented as clinically validated until it has appropriate external/prospective validation.